

Mohssine ASSEL

AI Engineer | Machine Learning | Generative AI

Available immediately

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PROFILE

Artificial Intelligence Engineer specializing in **Generative AI, Machine Learning, and NLP**. Experienced in designing and developing solutions based on **LLMs, RAG, AI agents, and Machine Learning models**. Proficient in the full AI development lifecycle, from data preparation to evaluation, productionization, and API integration. Able to turn business challenges into robust, operational AI solutions.

EDUCATION

Master's Degree – Information Systems and Decision Support

ISIMA

2025 – 2026 Clermont-Ferrand, France

Engineering Degree – AI and Computer Engineering

ENSAM Casablanca

2020 – 2025 Casablanca, Morocco

SKILLS

AI & ML

Python · PyTorch · Hugging Face · Transformers · Deep Learning · Computer Vision

Generative AI

LLMs · RAG · Prompt Engineering · Fine-tuning · Embeddings · AI Agents

Backend & Data

FastAPI · MongoDB · Data Pipelines · FAISS

MLOps & Cloud

Docker · Airflow · GitHub Actions · Model Deployment · Azure

Tools

Git · Linux · LangChain · LlamaIndex · DeepEval · OpenCV · Pydantic

LANGUAGES

Arabic	Native
French	Fluent
English	Fluent
Spanish	Basic

CERTIFICATIONS

Machine Learning with Python
IBM

SQL (Advanced)
HackerRank

PROFESSIONAL EXPERIENCE

Data Scientist

Crédit Agricole CIB – IGE-MS — Apr. 2026 – Oct. 2026 — Montrouge, France

- Designed and developed an NLP fuzzy-matching engine combining lexical similarity, distance algorithms, text normalization, and synonym handling to automate the detection of matches between texts.
- Developed, trained, and evaluated Machine Learning and NLP models for text classification, topic modeling, and sentiment analysis on business data.
- Implemented preprocessing pipelines, evaluation metrics, and explainable AI (XAI) techniques to improve model reliability and facilitate integration into business processes.

Technologies: Python, Dataiku, Scikit-learn, Sentence-Transformers, Embeddings

AI Engineer

BMCI – BNP Paribas Group — Feb. 2025 – Aug. 2025 — Casablanca, Morocco

- Developed a multilingual embedding-based system to find similar incidents, suggest resolutions, and predict incident priority and category through a FastAPI service.
- Productionized the MLOps pipeline using Airflow and MLflow, including hourly data ingestion, data quality checks, weekly model retraining, and model evaluation before deployment.

Technologies: Airflow, MLflow, PostgreSQL, Sentence-Transformers, FastAPI, Docker

AI Engineer

3D Smart Factory — Feb. 2024 – Jul. 2024 — Remote

- Developed a RAG system for answering questions in French and English about scientific papers, providing precise citations (paper, section, and page) and using OCR to process scanned documents.
- Designed a hybrid retrieval pipeline combining embeddings, BM25, RRF, and cross-encoder reranking, improving Hit@5 from 0.88 to 1.00 and MRR@5 from 0.78 to 0.94.

Technologies: Ollama, ChromaDB, PyMuPDF, Tesseract, FastAPI, Streamlit, Docker, Pydantic

Python Developer

Toyota — Jul. 2023 – Sep. 2023 — Casablanca, Morocco

- Analyzed and prepared production data, monitored industrial KPIs, and developed an interactive Power BI dashboard with automated data processing to improve production performance monitoring.

Technologies: Power BI, pandas

PROJECTS

RAG Evaluation Framework

- Developed a RAG agent and evaluation framework to measure answer quality, retrieval accuracy (Hit Rate, MRR), and hallucination rate, while comparing LLM providers (OpenAI, Gemini, Groq) and embedding models.

Technologies: Python, LangChain, Pydantic, DeepEval

Marathon Photo Retrieval System

- Developed an automated bib-number recognition system for marathon photos, enabling participants to easily find their own pictures using Computer Vision and OCR.

Technologies: Python, OpenCV, YOLO, OCR, PyTorch